MIL-S-28904(MC)
20 December 1968
SUPERSEDING
See Section 6

MILITARY SPECIFICATION

STRAINER ASSEMBLY, SKID MOUNTED

1. SCOPE

1.1 This specification covers requirements for one type, 4-inch skid mounted strainer assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-A-200/8	- Aluminum Alloy Bar, Rod, Shapes Tube and Wire, Extruded, and Structural, Shapes, 6061.
QQ-A-250/11	- Aluminum Alloy 6061, Plate and Sheet.
QQ-A-601	- Aluminum-Alloy Sand Castings.
QQ-P-416	- Plating, Cadmium (Electrodeposited).
RR-C-271	- Chains and Attachments, Welded, Weldless, and Roller Chain.
TT-E-527	- Enamel, Alkyd, Lusterless.
VV-F-800	- Fuel Oil, Diesel.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-T-60	- Tape, Pressure-Sensitive Adhesive, Waterproof, for Packaging.

MILITARY

MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-G-3056	- Gasoline, Automotive, Combat.
MIL-G-5572	- Gasoline, Aviation, Grades 80/87, 100/130, 115/145.
MIL-T-5624	- Turbine Fuel, Aviation Grades, JP-4 and JP-5.

FSC 4930

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MIL-C-6183	- Cork and Rubber Composition Sheet, for Aromatic Fuel and Oil Resistant Gaskets.
MIL-W-8604	- Welding of Aluminum Alloys; Process for.
MIL-A-8625	- Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-L-21260	- Lubricating Oil, Internal-Combustion Engine, Preservative.
MIL-C-27487	- Coupling Halves, Quick Disconnect-Cam-Locking Type.
MIL-V-58039	- Valves, Gate, Rising Stem, Double Acting, Aluminum.

STANDARDS

MILITARY

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing with Appropriate Test Methods.
MS33586	- Metals, Definition of Dissimilar.

DRAWINGS

MARINE CORPS

4115-4-1	- Wye Fitting, Flanged.
4115-5	- Valve Gate, 4-inch.
4117-1	- Gaskets, Rubber and Cork.
4127	- Lead, Electrical Grounding.
4129	- Strainer Assembly, Skid Mounted.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402).

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), as soon as practicable after award of contract or order, and prior to submission of any complete item for acceptance, the contractor shall furnish a first article for determination of conformance to this specification. The first article shall be representative of the product which the manufacturer will subsequently produce in fulfillment of the contract. Approval of the first article by the activity concerned shall not relieve the contractor of the obligation to supply strainer assemblies conforming to this specification. Any changes or deviations of the production items from the first article shall be subject to approval by the contracting officer.

3.2 Materials. Materials shall be as specified herein and on Drawing No. 4129. Any material not definitely specified shall be submitted to the contracting officer for approval.

3.2.1 Fuel resistance. All materials used for fabrication of strainer assemblies shall be resistant to the fuels specified in VV-F-800, MIL-G-3056, MIL-G-5572 and MIL-T-5624.

3.2.2 Metals. Metal parts shall be corrosion-resistant or shall be protected against corrosion by means of chemical or electrolytic finishes and treatments, or a combination thereof. Steel components shall be cadmium plated in accordance with type II class 1, of QQ-P-416.

3.2.2.1 Dissimilar metals. Contact between dissimilar metals shall be avoided whenever practical. Wherever it is necessary that dissimilar metals be assembled in intimate contact with each other, an interposing, insulating or mutually compatible material shall be used. Dissimilar metals are defined in MS33586. The following exceptions will be permitted:

(a) Bronze cam arms and bronze or stainless steel pins in quick disconnect coupler fittings.

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(b) Brass or bronze internal working parts.

3.2.2.2 Aluminum alloy sand castings. Aluminum alloy sand castings shall conform to class 3M, condition T6 or class 17M, condition T5 of QQ-A-601. Castings shall be of high quality, clean, sound and free from porosity, blow holes, cracks, and any other defects.

3.2.2.3 Plate and sheet aluminum alloy. Plate and sheet aluminum alloy shall conform to QQ-A-250/11 (6061 temper T6).

3.2.2.4 Aluminum bars, rods, and shapes. Aluminum bars, rods, and shapes shall conform to QQ-A-200/8 (6061 temper T6).

3.2.2.5 Security chain. Chain for securing dust caps and plugs to the associated fittings shall conform to class 3, type II of RR-C-271, and shall be equipped with two heavy duty bronze S-hooks, (one at each end). Both the chain and hooks shall be cadmium plated in accordance with QQ-P-416.

3.2.3 Gaskets. Flange gaskets shall be synthetic rubber and cork composition conforming to class 1, medium grade of MIL-C-6183 and Drawing No. 4117-1. Synthetic rubber gaskets for the quick disconnect coupler fittings and dust caps shall be in accordance with MIL-C-27487.

3.2.4 Fungus inert material. Materials which provide a nutrient medium for fungus and insects shall not be used.

3.3 Design and construction. The strainer assembly shall be a completely self-contained, skid mounted unit in a rollover aluminum pipe frame equipped with two 350 gallon per minute sediment strainers, four rising stem gate valves, one 4-inch inlet and one 4-inch outlet quick disconnect fitting.

3.3.1 Sediment line strainers. There shall be two sediment line strainers of 4-inch size which shall be of the top outlet type. Inlet and outlet ends shall have standard truck type flanges, faced and drilled. The body shall have a tapped plug in the top section for air relief and a bottom drain plug. The top shall have a removable cap for removing the strainer cage. The strainer cage shall be cadmium plated steel with a 40-mesh stainless steel basket. The strainer shall be aluminum.

3.3.2 Coupling. Each strainer assembly shall be equipped with two quick disconnect coupling halves in accordance with

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MIL-C-27487, one female at the inlet and one male at the outlet as shown on Drawing No. 4129. A dust cap and plug conforming to MIL-C-27487 shall be furnished with each fitting as appropriate. Each cap and plug shall be secured with a chain. All couplings shall provide a tight seal with mating fittings.

3.3.3 Valves. Each line strainer shall have a 4-inch gate valve, in accordance with MIL-V-58039 and Drawing No. 4115-5 at each end as shown on Drawing No. 4129. The gate valves shall be connected to the inlet and outlet quick disconnect fitting by an aluminum flanged end elbow and a flanged Y fitting as provided for on Drawing No. 4115-4-1.

3.3.4 Skid and frame assembly. The skid and frame assembly shall be of all aluminum welded construction in accordance with Drawing No. 4129. Frame shall be the rollover type, and of sufficient height and length to provide ample protection from damage to the valve stems and quick disconnect fittings. Towing eyes with a minimum inside diameter of 3 inches shall be provided on each end of the skids. Two sections of tubing with thumb screws shall be provided for securing the ground rod when not in use. Lifting eyes shall be provided on the rollover frame.

3.3.5 Ground rod and ground wire. A ground rod and ground wire, conforming to type I as shown on Drawing No. 4127 shall be provided. The ground rod shall be secured to the frame as provided for in accordance with 3.3.4 and the ground wire shall be wound around members of the skid frame when not in use.

3.3.6 Interchangeability. All parts having the same manufacturer's part numbers shall be physically and functionally interchangeable.

3.3.7 Welding. All aluminum welding shall be performed by the inert gas shielded arc method, in accordance with MIL-W-8604. The quality and strength of the welds shall be such that the assemblies will meet the requirements of this specification.

3.3.8 Special tools. No special tools shall be required for operation and maintenance of the strainer assemblies or components thereof.

3.3.9 Screw threads. All screw threads shall conform to the standards set forth in H-28.

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3.3.10 Identification marking. Identification marking shall be in accordance with Drawing No. 4129 and MIL-STD-130.

3.3.11 Nameplate. A U.S. nameplate shall be securely attached, in a conspicuous location, to each strainer assembly. These plates will be furnished upon request to the Commandant of the Marine Corps, Code CSG, Washington, D. C. 20380. The contractor shall legibly imprint the required data in the blank spaces as directed by the contracting officer.

3.3.12 Cleaning. All parts shall be clean and free from lint, sand, metal chips and weld splashes.

3.4 Performance and product characteristics.

3.4.1 Lifting. The strainer assembly shall be capable of being hoisted by means of the lifting eyes without permanent deformation or damage to any part when subjected to the test specified in 4.6.2.

3.4.2 Hydrostatic pressure. When tested as specified in 4.6.3 each strainer assembly shall be capable of withstanding a hydrostatic pressure of 200 pounds per square inch (psi).

3.4.3 Towing. When tested as specified in 4.6.4, the strainer assembly, completely filled with test fluid, shall be capable of being towed by means of the towing eyes.

3.5 Finish.

3.5.1 Anodizing. Aluminum parts not painted shall be color anodized in accordance with MIL-A-8625; the color shall match Marine Corps green No. 23, lusterless (see 6.3).

3.5.2 Treatment and painting. All items not anodized shall be treated and painted to match Marine Corps green No. 23, lusterless (see 6.3), in accordance with type A of MIL-T-704, except that the finished coat of paint shall conform to TT-E-527.

3.6. Workmanship. Each strainer assembly shall be manufactured in a careful and workmanlike manner in accordance with sound practice consistent with the current state of the art.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 General inspection and test.

4.2.1 Test conditions. Tests specified herein may be performed under any natural combination of prevailing ambient conditions. All test conditions and the results of each test shall be observed and recorded.

4.2.2 Test fluid. The test fluid shall be any type of automotive fuel locally available in quantity. Water shall not be used for any test specified herein, except the hydrostatic test.

4.3 Classification of tests. Inspection and testing shall be classified as follows:

(a) First article tests (see 4.4 and 6.4).

(b) Acceptance tests (see 4.5).

(c) Preparation for delivery (see 4.7).

4.4 First article tests. A completely fabricated strainer assembly shall be subjected to all the inspections and tests specified in 4.6.1 through 4.7.3. These tests shall be made by the contractor in the presence of a Government inspector for approval of the contracting officer prior to any acceptance by the Government (see 6.4).

4.5 Acceptance tests. Each strainer assembly submitted for acceptance shall be subjected to the inspections and tests specified in 4.6.1 and 4.6.3. These tests shall be performed by the contractor in the presence of a Government inspector. Failure to pass any test shall be cause for rejection.

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4.6 Test procedures. The following detailed examination and test procedures shall be used for determination of compliance with the requirements specified herein.

4.6.1 Visual and mechanical inspection. Strainer assemblies shall be inspected to verify that the materials, design, workmanship, dimensions, weight, marking and construction conform to the requirements of this specification.

4.6.2 Lifting. The strainer assembly, ballasted to an evenly distributed gross weight of 500 pounds minimum, shall be hoisted so that the entire weight of the assembly and ballast is supported by the two lifting eyes for a minimum period of 10 minutes. During and after this test, the entire assembly, especially the lifting eyes and frame, shall be inspected for evidence of permanent deformation or failure. The strainer assembly shall then be subjected to the tow test specified in 4.6.4.

4.6.3 Hydrostatic pressure test. Strainer assemblies shall be hydrostatically tested. Each line strainer shall be tested separately as follows:

(a) With the cap on the outlet fitting, the two valves open on one line strainer and closed on the other line s strainer, connect the inlet fitting to a source of test fluid.

(b) Fill strainer assembly with test fluid being careful to eliminate entrapped air through the plug in the top of the line strainer body.

(c) Raise the inlet pressure to 200 psi and maintain for a minimum period of 5 minutes.

(d) During the test, observe the assembly for leakage and any evidence of failure.

(e) Repeat tests (a) through (d) for the second line strainer.

4.6.4 Towing test. With all valves open, the dust cap and plug in place in the inlet and outlet fittings and the assembly completely filled with test fluid, the assembly shall be towed by means of the towing eyes for a minimum distance of 400 feet across a hard surfaced road at a minimum speed of 5 miles per hour (mph). After towing, the assembly shall be subjected to the following inspection and tests:

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(a) Visual inspection for major damage and leakage. Nicks, dents, scrapes and marred finishes are not to be considered major damage.

(b) Hydrostatic pressure specified in 4.6.3.

4.7 Packaging and packing tests. A pilot pack shall be subjected to the following tests. The pack shall contain the assembly and accessories secured in place and ready for shipment.

4.7.1 Impact test. The crate, suspended by means of slings, or grabhooks, shall be raised to a height sufficient to allow the crate to swing clear of the ground. The crate shall be swung to cause one end or corner to strike against a solid barrier at a speed of at least 8 mph. The test shall be repeated for the opposite end of the crate.

4.7.2 Drop test. One end of the crate shall be mounted on a wood sill a minimum of 6 inches high placed at right angles to the skids. The opposite end shall then be raised to a height of 36 inches and allowed to fall free to a hard surface. This test shall be applied once to each end.

4.7.3 Examination. After completion of the above tests, the pack shall be examined for deficiencies, which shall be corrected by redesign and the crate retested.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. After completion of acceptance testing the strainer assembly shall be completely filled with preservative oil, conforming to grade 2 of MIL-L-21260. Oil shall be retained in the assembly for 1 hour after which the assembly shall be completely drained and the dust caps and plugs installed. Valves shall be opened to a minimum of 1/4 inch. Valves and end fittings shall be wrapped with barrier material conforming to grade C of MIL-B-121 and sealed with pressure sensitive tape conforming to PPP-T-60.

5.1.2 Level C. The strainer shall be preserved and packaged in accordance with the manufacturer's standard practice to afford protection against deterioration and damage.

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5.2 Packing. Packing shall be level A or C as specified (see 6.2).

5.2.1 Level A. Each strainer shall be packed in a container conforming to class 2 of PPP-B-621, or overseas type, style A or B, plywood, type III, class 1 of PPP-B-601. Containers shall be modified by the addition of not less than two nominal 2" by 4-inch wood skids positioned flatwise across the bottom of the box. The ends of the skids shall be beveled to full depth at a 45° angle. Blocking, bracing, and anchoring of the assembly within the container shall be in accordance with MIL-STD-1186.

5.2.2 Level C. Each strainer assembly shall be packed in a manner to insure carrier acceptance and safe delivery to destination in containers complying with the carrier rules and regulations applicable to the mode of transportation.

5.3 Marking. Shipments shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The strainer assembly functions to remove solid contaminates from liquid petroleum fuels.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Whether a first article is required (see 3.1).

(c) Levels of preservation, packaging and packing (see 5.1 and 5.2).

6.3 Color. Color chips for Marine Corps green No. 23 lusterless may be obtained from the Commandant of the Marine Corps (Code CSY-10), Headquarters, U.S. Marine Corps, Washington, D. C. 20380.

6.4 First article. The first article shall be examined and tested for approval at the contractor's plant or at an independent commercial testing laboratory acceptable to the procuring activity. First article tests shall be witnessed by a representative of the procuring activity. Approval of the first article shall be by the contracting officer.

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6.5 Supersession data. This specification includes the requirements of Marine Corps Purchase Description HQPD-64-26(2) Strainer Assembly, Skid Mounted, dated 6 September 1967.

Preparing activity:
Navy - MC
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